REGULAR, KOSZUL REGULAR, H1-REGULAR AND QUASI-REGULAR SEQUENCES AGREE OVER A NOETHERIAN LOCAL RING

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Allowed standard prerequisites (editorial).

- Regular implies Koszul regular and Koszul regular implies H1-regular. Lemma \ref{lemma-regular-koszul-regular} and Lemma \ref{lemma-koszul-regular-H1-regular}, cited at proof lines 7461 and 7462-7463. Give the first two implications.
- H1-regular implies quasi-regular. Lemma \ref{lemma-H1-regular-quasi-regular}, cited at proof lines 7464-7465. Gives the third implication.
- Quasi-regular implies regular over a Noetherian local ring. Algebra, Lemma \ref{algebra-lemma-quasi-regular-regular}, cited at proof lines 7466-7467. Closes the cycle.

Source reference. Original lemma 09CC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11825–11825. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 062F: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7794–7794. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0CEM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15582–15582. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 062I: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7797–7797. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 061S: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7771–7771. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $(R, \mathfrak{m})$ be a Noetherian local ring. Let M be a nonzero finite R-module. Let $f_1, \ldots, f_r \in \mathfrak{m}$. The following are equivalent*

(1) *$f_1, \ldots, f_r$ is an M-regular sequence,*
(2) *$f_1, \ldots, f_r$ is a M-Koszul-regular sequence,*
(3) *$f_1, \ldots, f_r$ is an M-H_1-regular sequence,*
(4) *$f_1, \ldots, f_r$ is an M-quasi-regular sequence.*

In particular the sequence $f_1, \ldots, f_r$ is a regular sequence in R if and only if it is a Koszul regular sequence, if and only if it is a H_1-regular sequence, if and only if it is a quasi-regular sequence.

Proof. The implication (1) $\Rightarrow$ (2) is Lemma 062F. The implication (2) $\Rightarrow$ (3) is Lemma 0CEM. The implication (3) $\Rightarrow$ (4) is Lemma 062I. The implication (4) $\Rightarrow$ (1) is Algebra, Lemma 061S. $\square$